

UNIT – 4
ASSESSMENT – 1
PROBLEMS

NIHARIKA PREEJESH 192524506

TASK 1 — DATA PREPARATION & INDUCTIVE LEARNING

PROBLEM

A healthcare company has collected patient records containing **Age, Blood Pressure, Cholesterol, Glucose Level, BMI and Family History** to develop an AI system for early diabetes diagnosis.

The task is to:

1. Apply an inductive learning approach.
2. Identify the target variable and important features.
3. Handle class imbalance between diabetic and non-diabetic patients.
4. Split the dataset into **70% training and 30% testing** data.
5. Apply suitable preprocessing such as normalization and encoding.

TASK 2 — DECISION TREE FOR DIAGNOSIS

PROBLEM

Develop a **Decision Tree classifier** to predict whether a patient is diabetic.

The system must:

1. Build a Decision Tree.
2. Determine the root node using Gini Index or Information Gain.
3. Display the first three levels of the tree.

4. Calculate Accuracy, Precision, Recall and F1-score.
5. Identify False Negatives.
6. Compare pre-pruning and post-pruning.

TASK 3 — STATISTICAL LEARNING FOR RISK STRATIFICATION

PROBLEM

Apply a statistical learning model to the same diabetes dataset and compare it with the Decision Tree.

The system must:

1. Use **Logistic Regression**.
2. Calculate Accuracy and AUC-ROC.
3. Perform feature importance analysis.
4. Identify the top three predictors.
5. Compare the predictors obtained from both models.

TASK 4 — REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

PROBLEM

The healthcare company wants an AI agent that recommends a suitable action based on a patient's health state.

The possible actions are:

Diet

Exercise

Medication

Monitor

The task is to:

1. Model the problem as an MDP.
2. Define states, actions, rewards and transitions.
3. Apply Q-Learning for at least five episodes.
4. Display Q-table updates.
5. Extract the final policy.
6. Discuss safety, bias and explainability.